Classical Field Theory — Exercise Sheet 8

Slices, boundary conditions and causal brackets

Solutions

For real Klein–Gordon variations with compact smooth Cauchy data, use $\omega^\mu(h_1, h_2) = h_1\partial^\mu h_2 - h_2\partial^\mu h_1$. The future-oriented graph element is $(1, -\nabla\tau)\mathrm{d}^d x$. All fields below are smooth. The last problem explicitly restricts the field equation to the massless wave in $3+1$ dimensions.

Exercise 8.1 (An unbounded Cauchy graph). Let $\tau(x) = t_* + v \cdot x$ on $\mathbb{R}^d$, where $t_* \in \mathbb{R}$ and $v \in \mathbb{R}^d$, $|v| < 1$. Let $h_1, h_2 \in \mathcal{S}_m$ be the actual compact-Cauchy-data solutions of Lecture 8, $m \geq 0$.

(a) Prove that the graph is spacelike and meets each inextendible timelike curve exactly once. A future timelike curve can be parametrized by coordinate time with $|\dot{x}(t)| < 1$; justify that its time range is all of $\mathbb{R}$. Here an inextendible curve has no past or future endpoint in Minkowski spacetime.

(b) Although τ is unbounded, prove that the current is supported in a compact subset of the region between the graph and $t = 0$. Give an explicit spatial bound in terms of t_*, $|v|$, and a radius R containing both sets of initial data. Use signed time integration to prove $\Omega_{\Sigma_\tau}(h_1, h_2) = \Omega_0(h_1, h_2)$, including where the graph crosses zero.

(c) Choose nonzero real $f \in C_c^\infty(\mathbb{R}^d)$ and solutions with data $(f, 0)$ and $(0, f)$. Determine their pairing on this graph.

For comparison, the following general result is supplied: in full Minkowski spacetime, the same compact-data solutions have finite, equal symplectic integrals on every smooth spacelike Cauchy surface with future orientation. It uses spacelike compact support, not just current closure. See Khavkine, §3.3.3, Lemma 3.11, p. 41, arXiv:1402.1282. Your proof here should establish the affine case directly.

Solution. A graph tangent $(v \cdot a, a)$ has Minkowski square $(v \cdot a)^2 - |a|^2 < 0$ for $a \neq 0$, so the graph is spacelike. On a future timelike curve parametrized by t,

$$\frac{\mathrm{d}}{\mathrm{d}t}(t - t_* - v \cdot x(t)) = 1 - v \cdot \dot{x}(t) \geq 1 - |v| > 0.$$

If the time range had a finite upper endpoint, the speed bound would make $x(t)$ Cauchy there. Its finite limit would allow a continuation by a timelike curve, contradicting inextendibility. The same argument applies at the lower endpoint. Thus the time range is $\mathbb{R}$ and the displayed function runs from $-\infty$ to $+\infty$, with exactly one zero.

Finite propagation gives support of both solutions and their derivatives inside $|x| \leq R + |t|$. Between 0 and $\tau(x)$, one also has $|t| \leq |t_*| + |v||x|$. Where the current can be nonzero this implies

$$(1 - |v|)|x| \leq R + |t_*|.$$

The spatial bound also bounds t, so the relevant closed support in the comparison region is compact. In particular both graph and flat integrals are finite.

Define $A^i(x) = \int_0^{\tau(x)} \omega^i(t, x)\mathrm{d}t$ with its signed orientation. The support bound makes A^i vanish outside a fixed ball. The product rule, FTC and $\partial_\mu \omega^\mu = 0$ give

$$\omega^0(\tau(x), x) - \omega^i(\tau(x), x)v_i - \omega^0(0, x) = -\partial_i A^i(x).$$

Integrating in a large rectangle gives zero on the right by coordinate FTC. The left uses the future graph element $(1, -v)$, proving the equality even when τ changes sign. For the final data the flat pairing is $\int [f \cdot f - 0 \cdot 0] \mathrm{d}^d x = \|f\|_2^2 > 0$, so the same value holds on the affine graph. This direct proof did not assume the supplied general theorem.

Exercise 8.2 (Common boundary conditions remove side flux). Let two real solutions of $(\partial_t^2 - \partial_x^2 + m^2)h = 0$ be smooth on a neighbourhood of $I \times [0, L]$, where I is an open interval, $L > 0$ and $m \geq 0$. Set $\Omega_{[0,L]}(h_1, h_2) = \int_0^L (h_1 h_{2,t} - h_2 h_{1,t}) \mathrm{d}x$. At each endpoint, consider the following common conditions on both solutions: Dirichlet $h = 0$, Neumann $\partial_n h = 0$, or real Robin $\partial_n h + \kappa h = 0$, with the same prescribed real κ for both solutions at that endpoint. The two endpoints may have different such conditions.

(a) Prove that each choice makes the side symplectic flux zero and conserves $\Omega_{[0,L]}$. Specify the outward derivative at each endpoint.

(b) For $m = 0$, $L = \pi$, compare $h_D(t, x) = \sin x \cos t$ and $h_N(t, x) = 1$. Each obeys one of the listed conditions, but they obey different conditions. Compute their current, pairing and net outward side flux. Is individual satisfaction of some listed condition sufficient?

Solution. The outward derivatives are $\partial_n = -\partial_x$ at zero and $\partial_n = +\partial_x$ at L. Since $\omega^1 = -h_1 h_{2,x} + h_2 h_{1,x}$, its outward contraction at either endpoint is

$$n\omega^1 = -(h_1 \partial_n h_2 - h_2 \partial_n h_1).$$

Common Dirichlet conditions kill both undifferentiated factors. Common Neumann conditions kill both normal derivatives. For common Robin conditions, substituting $\partial_n h_j = -\kappa h_j$ makes the two terms equal and opposite. Thus each side flux is zero. Integrating the current divergence over the interval gives

$$\frac{\mathrm{d}}{\mathrm{d}t} \Omega_{[0,L]} = -[\omega^1(t, L) - \omega^1(t, 0)] = 0.$$

For the second part, h_D solves the wave equation and vanishes at both endpoints, while h_N solves it and has zero normal derivative. Their ordered current is

$$\omega^0 = \sin x \sin t, \qquad \omega^1 = \cos x \cos t.$$

Integration gives $\Omega = 2 \sin t$. The net outward flux is $\omega^1(\pi) - \omega^1(0) = -2 \cos t$, whose negative is the derivative of the pairing. A Dirichlet solution paired with a Neumann solution need not have zero boundary current. Common boundary conditions, rather than separate choices for each solution, supply the cancellation.

Exercise 8.3 (A causal source bracket and its sign). In $3+1$ dimensions let G_{ret} be Lecture 6's massless kernel, let $G_{\mathrm{adv}}(t,x) = G_{\mathrm{ret}}(-t,x)$, and put $E = G_{\mathrm{adv}} - G_{\mathrm{ret}}$. For real $f, g \in C_c^\infty(\mathbb{R}^{1+3})$, define on the massless compact-data solution space

$$F_f(\phi) = \int_{\mathbb{R}^{1+3}} f\phi \, \mathrm{d}t\mathrm{d}^3x, \qquad Ef = (G_{\mathrm{adv}} - G_{\mathrm{ret}}) * f.$$

The identification of the resulting causal bracket with the linear *Peierls bracket* may be taken as given; see Khavkine, §§3.3.4–3.3.5, Theorem 3.2, and the scalar case §4.1, arXiv:1402.1282. The signs in this problem must be fixed by the following calculation, using the course convention $\iota_{X_F}\Omega = DF$.

(a) Show that Ef is a smooth homogeneous solution with compact smooth data on every constant slice. Apply the sourced current balance to $G_{\mathrm{ret}} * f$ and a homogeneous compact-data solution h to prove $\Omega(Ef, h) = \int fh$. Identify X_{F_f} and derive the bracket $\{F_f, F_g\} = \int f(Eg)$. Prove its antisymmetry and its vanishing when the two test supports are spacelike separated.

(b) As a separate finite-dimensional sign test, take one oscillator $\ddot{q} + \nu^2 q = 0$, $\nu \geq 0$, with $\{q(0), p(0)\} = 1$ and $p = \dot{q}$. Compute $\{q(t), q(s)\}$ and $\{q(t), p(s)\}$, using continuous limits at $\nu = 0$. Recover the equal-time canonical sign. Explain why this finite model does not make point evaluation of a field admissible in Lecture 7.

Solution. Both convolutions are smooth sourced solutions by Lecture 6 and time reflection. Their difference is homogeneous. If the source lies in $[a,b] \times B_R$, the retarded contribution on a fixed slice is spatially supported in a ball of radius at most $R + \max(t-a, 0)$ and is zero for $t < a$. The advanced counterpart has the analogous radius $R + \max(b-t, 0)$ and is zero for $t > b$. The same is true of all derivatives. Thus Ef has compact smooth Cauchy data and belongs to the actual tangent space.

Let $r_f = G_{\mathrm{ret}} * f$ and let h be homogeneous. The massless current satisfies

$$\partial_\mu \omega^\mu(r_f, h) = r_f \Box h - h \Box r_f = -hf.$$

Choose constant slices $t_- < a$ and $t_+ > b$. Compact support on this bounded time interval removes every side term in a sufficiently large rectangle. The initial current is zero. Its integrated balance is

$$\Omega_{t_+}(r_f, h) = -\int fh \, \mathrm{d}t\mathrm{d}^3x.$$

At the late slice the advanced field is zero, so $Ef = -r_f$. Lecture 8's slice independence for the homogeneous pair then gives $\Omega(Ef, h) = \int fh = DF_f(h)$ on every constant slice. Nondegeneracy identifies $X_{F_f} = Ef$. The bracket order is therefore

$$\{F_f, F_g\} = DF_f(X_{F_g}) = \int f(Eg) = \Omega(Ef, Eg) = -\Omega(Eg, Ef) = -\{F_g, F_f\}.$$

If every event in the first compact support is spacelike separated from every event in the second, the retarded and advanced support properties make Eg zero on the support of f. The bracket vanishes. This proves the displayed linear bracket identity; the supplied Peierls identification is not a theorem here about arbitrary nonlinear theories.

For the finite oscillator, write

$$q(t) = \cos(\nu t) q(0) + \frac{\sin(\nu t)}{\nu} p(0).$$

The canonical two-variable bracket gives

$$\{q(t), q(s)\} = \frac{\cos(\nu t)\sin(\nu s) - \sin(\nu t)\cos(\nu s)}{\nu} = \frac{\sin(\nu(s-t))}{\nu}.$$

Differentiating with respect to s gives $\{q(t), p(s)\} = \cos(\nu(s-t))$. For $\nu = 0$ the solution is $q(0)+tp(0)$ and the same brackets are $s-t$ and 1. This agrees with $E(t-s) = -\sin(\nu(t-s))/\nu$ for the advanced-minus-retarded oscillator response. Here $q(t)$ is a smooth function of two real coordinates. Field point evaluation instead has a delta derivative, excluded from the Schwartz-gradient class of Lecture 7.